Sahaj Bhadja

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Summary

Software engineer with 2+ years building **Python/C++-adjacent systems**, **secure APIs**, and **cloud workflows** across healthcare, autonomy research, and product environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected **Azure SQL and Blob services** for 500+ users, tuning indexed queries to cut dashboard latency from **1.2 seconds to 350 milliseconds**.
- Built **fault-tolerant job execution** for 1,000+ monthly workflows with retries and isolation, improving throughput by **3x** under concurrent production workloads.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered **Flask APIs** and React workflows handling **10,000+ daily requests**, integrating secure ingestion and AI-assisted grading for production educational operations.
- Designed **multithreaded processing** for 1,000+ uploads on AWS RDS-backed services, reducing feedback turnaround by **60%** while improving throughput and debugging.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **transformer evaluation pipelines** across **100K+ trajectory logs**, enabling reproducible experimentation and high-confidence benchmarking for long-horizon model behavior.
- Architected **map-aware encoders** with spatial-temporal attention, improving closed-loop planning stability by **15%** while strengthening debugging rigor.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered **Python API and ETL microservices** with SQL, AWS Glue, and S3, improving response times by **30%** across distributed data workflows.
- Maintained **Jenkins CI pipelines** and 20+ automated suites with PyTest and Mockito, increasing release reliability by **68%** across shared environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built **MERN APIs and WebSocket modules** supporting **2M+ synchronized records**, improving consistency across real-time modular workflows.
- Automated **Docker-based CI/CD workflows** with Jenkins, cutting release time by **50%** while improving deployment consistency and troubleshooting speed.

Skills

Languages: **Python**, **C++**, Java, TypeScript, JavaScript, SQL, Bash; **Software & Systems**: Flask, FastAPI, React, REST APIs, WebSockets, microservices, multithreading, scheduling algorithms, debugging, performance optimization
Data, Cloud & Quality: PostgreSQL, MySQL, SQL Server, MongoDB, AWS, Azure, Azure SQL, AWS RDS, S3, Blob Storage, AWS Glue, ETL, indexing, query tuning, Git, Docker, Jenkins, CI/CD, PyTest, Mockito, automated testing, Agile/Scrum, logging, monitoring, Sentry, Linux

Projects

CPU-scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2*
- Implemented **FCFS, SJF, SRT, RR, priority scheduling, and MLQ** with interactive controls, EJ-2 Gantt visualization, and dynamic waiting-time calculations across 1K+ simulated process combinations.

HMIS Live Live | *MERN, ABDM APIs, TypeScript, Twilio, Sentry*
- Built a **healthcare workflow platform** with ABDM APIs, Twilio alerts, and Sentry monitoring, enabling consent-based patient-history access and reducing administrative workload by **30%**.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered SQL workspace** for **10+ databases**, accelerating query authoring and schema exploration through unified Java and TypeScript services.